

COMPANION
to

ChatGPT Teaches TikZ

Chapter 5: Shape and Color Style (B)

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5.1 What This Companion Adds

Chapter 5 added shape refinement and color to the line-style vocabulary. This Companion develops a compact visual system based on rounded corners, outline color, fill color, and restrained color mixtures.

The goal is not to decorate every object differently. It is to make objects with the same role look alike and to reserve contrast for meaningful emphasis.

Design

A diagram should remain understandable when printed in grayscale or viewed at a small size. Color supports structure; it should not replace structure.

5.2 A Shape-and-Color Vocabulary

Option	Purpose
<code>rounded corners</code>	Rounds polygonal corners by a standard amount.
<code>rounded corners=6pt</code>	Sets the amount of rounding.
<code>draw=blue</code>	Sets the outline color.
<code>fill=yellow</code>	Sets the interior color.
<code>blue!20</code>	Mixes blue with white to make a light tint.
<code>blue!60!black</code>	Mixes blue with black to make a darker tone.

These options may be applied directly or collected in a reusable style.

5.3 Rounded Corners

Rounded corners change appearance without changing the vertices of a path.

```
\begin{tikzpicture}
  \draw (0,0) rectangle (3,1.4);
  \draw[rounded corners]
```

```
(4,0) rectangle (7,1.4);
\end{tikzpicture}
```

See



The same four corner coordinates are used in both rectangles.

5.3.1 Controlling the Amount

```
\draw[rounded corners=2pt]
(0,0) rectangle (3,1.4);

\draw[rounded corners=10pt]
(4,0) rectangle (7,1.4);
```

Small values soften the corners slightly; large values make the rounding more conspicuous. Use one standard value for objects with the same role.

5.3.2 Polygonal Paths

The option also rounds turns in a multi-segment path.

```
\begin{tikzpicture}
\draw[rounded corners=6pt]
(0,0) -- (2,0) -- (2,1)
-- (4,1) -- (4,0);
\end{tikzpicture}
```

Common Mistake

The option is **rounded corners**, with a final **s**. It changes corners that already exist; it does not turn an arbitrary open path into a closed shape.

5.4 Outline Color

Use **draw=** to choose the color of a path or node border.

```
\begin{tikzpicture}
  \draw[draw=blue]
    (0,0) rectangle (3,1.4);

  \node[draw=red] at (5,0.7) {Warning};
\end{tikzpicture}
```

The option **draw** by itself requests a border in the current drawing color. The option **draw=blue** both requests the border and chooses blue.

5.5 Fill Color

Use **fill=** for the interior of a closed path or node.

```
\begin{tikzpicture}
  \draw[draw=black,fill=yellow]
    (0,0) rectangle (3,1.4);

  \node[draw,fill=green!20]
    at (5,0.7) {Ready};
\end{tikzpicture}
```

See



Ready

An open line has no interior. Applying `fill=` to a single open segment therefore does not create a visible colored region.

5.5.1 Outline and Fill Together

Outline and fill are independent:

```
\node[
  draw=blue!60!black,
  fill=blue!15
] at (0,0) {Theorem};
```

A dark outline with a light tint usually preserves legibility better than a strong saturated fill.

5.6 Color Mixing

TikZ creates tints and mixtures with exclamation marks.

- `blue!10` is a very light blue.
- `blue!30` contains more blue.
- `blue!60!black` is a darker mixture.

```
\begin{tikzpicture}
  \node[draw=blue!60!black,fill=blue!10]
    at (0,0) {Light};
  \node[draw=blue!60!black,fill=blue!30]
    at (2.5,0) {Medium};
  \node[draw=blue!60!black,fill=blue!50]
    at (5,0) {Strong};
\end{tikzpicture}
```

See

Light

Medium

Strong

Use nearby tints of one color when objects belong to the same visual family.

5.7 Semantic Shape Styles

Repeated shape and color options belong in a style.

```
\tikzset{
  stage node/.style={
    draw=blue!60!black,
    fill=blue!10,
    rounded corners
  },
  focus node/.style={
    draw=red!70!black,
    fill=red!10,
    very thick,
    rounded corners
  }
}
```

The names describe roles. The definitions record the visual choices for those roles.

When two objects have the same role, give them the same style. When one object has a different role, change only enough visual properties to make that difference clear.

5.8 Worked Project: A Styled Process

The project begins with four named nodes and then applies one ordinary style and one emphasis style.

5.8.1 Stage 1: Define the Styles

```
\tikzset{
  stage node/.style={
    draw=blue!60!black,
    fill=blue!10,
    rounded corners
  },
  check node/.style={
    draw=red!70!black,
    fill=red!10,
    very thick,
    rounded corners
  }
}
```

5.8.2 Stage 2: Assign the Roles

```
\node[stage node] (input) at (0,0) {Input};
\node[stage node] (prepare) at (2.5,0) {Prepare};
\node[check node] (check) at (5,0) {Check};
\node[stage node] (output) at (7.5,0) {Output};
```

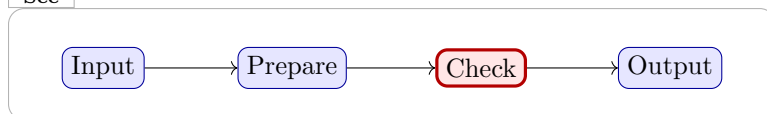

5.8.3 Stage 3: Connect the Nodes

```
\draw[->] (input) -- (prepare);  
\draw[->] (prepare) -- (check);  
\draw[->] (check) -- (output);
```

5.8.4 The Complete Picture

```
\begin{tikzpicture}  
  \tikzset{  
    stage node/.style={  
      draw=blue!60!black,  
      fill=blue!10,  
      rounded corners  
    },  
    check node/.style={  
      draw=red!70!black,  
      fill=red!10,  
      very thick,  
      rounded corners  
    }  
  }  
  
  \node[stage node] (input) at (0,0) {Input};  
  \node[stage node] (prepare) at (2.5,0) {Prepare};  
  \node[check node] (check) at (5,0) {Check};  
  \node[stage node] (output) at (7.5,0) {Output};  
  
  \draw[->] (input) -- (prepare);  
  \draw[->] (prepare) -- (check);  
  \draw[->] (check) -- (output);  
\end{tikzpicture}
```

See



Design

Only the checking stage uses the emphasis style. The colors, line weight, and rounded corners work together to show roles without changing the diagram's structure.

5.9 Debugging Workshop

5.9.1 Forgetting a Comma

Incorrect:

```
\node[draw=blue fill=yellow] at (0,0) {A};
```

Correct:

```
\node[draw=blue,fill=yellow] at (0,0) {A};
```

5.9.2 Confusing Outline and Fill

`draw=blue` changes the border. `fill=blue` changes the interior. Use both when both properties matter.

5.9.3 Trying to Fill an Open Segment

```
\draw[fill=yellow] (0,0) -- (4,0);
```

The path has no enclosed region, so there is nothing visible to fill.

5.9.4 Using Too Many Colors

If every node has a different color, readers cannot tell which differences are meaningful. Begin with one neutral family and one highlight family.

5.10 Exercises

1. Draw one ordinary rectangle and one rounded rectangle.
2. Compare `rounded corners=2pt` and `rounded corners=10pt`.
3. Draw a rectangle with a blue outline and no fill.
4. Draw a node with a black outline and yellow fill.
5. Compare three tints of one color.
6. Create a node with a dark blue outline, light blue fill, and rounded corners.
7. Define one ordinary node style and one emphasis style.
8. Restyle the worked project so that **Prepare**, rather than **Check**, receives emphasis.

5.11 Solutions

5.11.1 Exercises 1–4

```
\draw (0,0) rectangle (3,1.4);  
\draw[rounded corners]  
  (4,0) rectangle (7,1.4);
```

```
\draw[rounded corners=2pt]  
  (0,0) rectangle (3,1.4);  
\draw[rounded corners=10pt]  
  (4,0) rectangle (7,1.4);
```

```
\draw[draw=blue]
  (0,0) rectangle (3,1.4);
```

```
\node[draw=black,fill=yellow]
  at (0,0) {Ready};
```

5.11.2 Exercises 5–7

```
\node[draw=blue!60!black,fill=blue!10] at (0,0) {A};
\node[draw=blue!60!black,fill=blue!30] at (2,0) {B};
\node[draw=blue!60!black,fill=blue!50] at (4,0) {C};
```

```
\node[
  draw=blue!60!black,
  fill=blue!10,
  rounded corners
] at (0,0) {Theorem};
```

```
\tikzset{
  ordinary node/.style={draw,fill=blue!10,rounded corners},
  emphasis node/.style={draw,fill=red!10,very thick,rounded corners}
}
```

For Exercise 8, assign `check node` to `Prepare` and `stage node` to `Check`; no coordinate or arrow needs to change.

5.12 ChatGPT Workshop

Ask ChatGPT to keep the structure fixed while evaluating the design choices.

1. Keep every coordinate unchanged and simplify this shape-and-color system to one ordinary style and one emphasis style.
2. Explain the difference between `draw=blue` and `fill=blue` in this node.
3. Turn these repeated outline, fill, and corner options into one semantic TikZ style.
4. Check the commas and color-mixture syntax in this option list. Do not redesign the picture.
5. Evaluate whether this diagram would remain understandable in grayscale and suggest the smallest necessary revision.

5.13 Final Checklist

Before continuing to Chapter 6, check that you can

1. apply and control rounded corners;
2. distinguish outline color from fill color;
3. recognize why an open path has no visible fill;
4. create light and dark mixtures of one color;
5. combine shape, color, and line options with commas; and
6. collect repeated appearance choices in semantic styles.